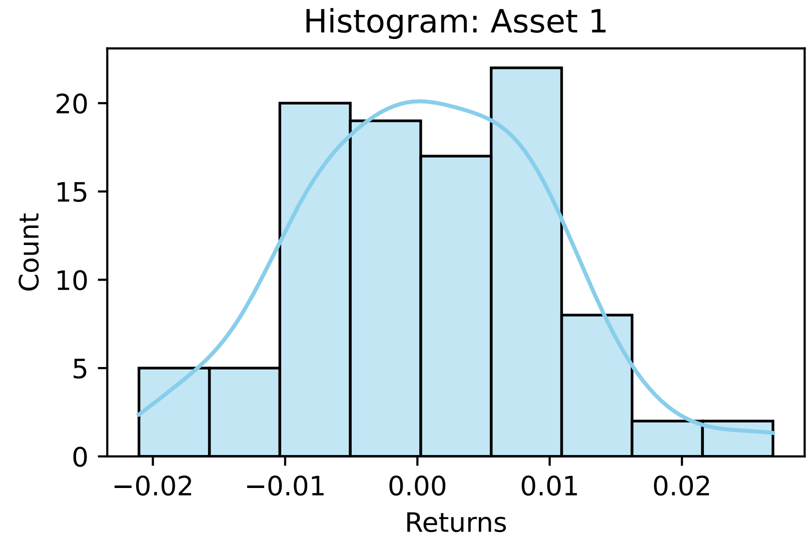
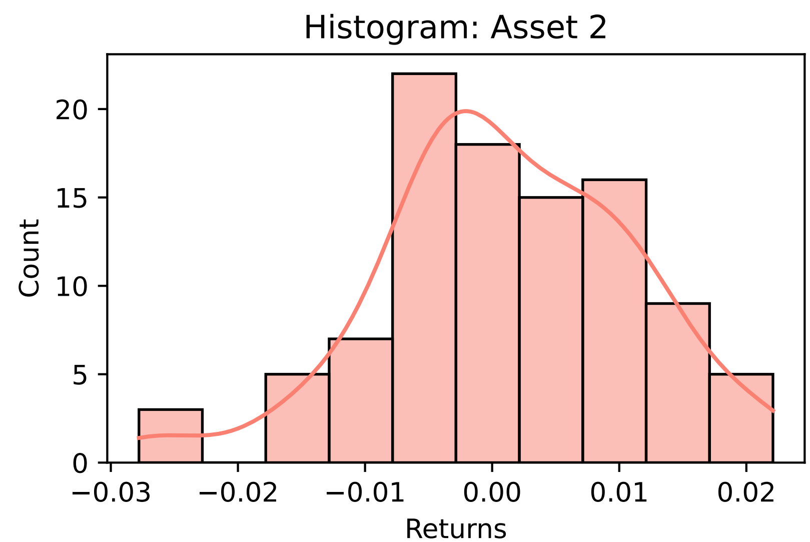
Data

Sample Size: 100 observations

Metric	Asset 1 (X)	Asset 2 (Y)
Count	100.0	100.0
Mean	0.000623	0.001033
Std	0.009447	0.010052
Min	-0.021052	-0.027786
25%	-0.00626	-0.004667
50%	0.000414	6.8e-05
75%	0.007137	0.00785
Max	0.026882	0.022093





Model, Method and Results

1. Model Formula:

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$$

2. Method:

- Nonparametric Bootstrap (B=1000 resamples)
- Explicit EDF Sampling: index = floor(n * U), U ~ Uniform(0,1)
- Confidence Interval: Percentile Method (2.5%, 97.5%)
- Libraries: pandas, numpy, matplotlib, seaborn

3. Model Assumptions:

- Returns are Independent and Identically Distributed (i.i.d).
- Sample represents a stationary data-generating process.

4. Results:

Alpha Hat: 0.5425 | 95% CI: [0.4086, 0.7075]

Visualization